Context-Aware Personalization and Adaptive Preference Elicitation in Conversational Recommender Systems

B.Tech Batch 2 – Research Project Plan | September–November 2026

Department of Computer Science and Engineering

Project orientation. This is a research-oriented project. The team will use an existing conversational recommendation dataset and published implementations, establish a reproducible baseline, introduce a focused personalization and preference-elicitation component, and evaluate the effect experimentally. A website, mobile application, chatbot interface, or other user-facing product is not required.

1. Problem Statement

Conversational Recommender Systems interact with users through multiple turns of conversation and use the information provided by the user to recommend suitable items. As a conversation progresses, the user's preferences may become clearer, change, or require further questioning.

The proposed project investigates whether a conversational recommender can maintain a useful representation of the user's current preferences, combine it with the user's available historical preferences, and use that information to decide whether to ask for more information or proceed with recommendation.

The primary dataset will be **TG-ReDial**, which provides multi-turn conversations together with user profiles and historical interaction information. ReDial may be used as a secondary validation dataset if time permits. The work will remain an offline research experiment rather than a user-facing application.

2. Recent Papers to be Studied

Li et al. (2018), “Towards Deep Conversational Recommendations.” Primary source for the ReDial setting and conversational recommendation.

Zhou et al. (2021), “CRSLab: An Open-Source Toolkit for Building Conversational Recommender System.” Useful for CRS implementations, datasets, model components, and evaluation.

Makarova et al. (2024), “Learning a Strategy for Preference Elicitation in Conversational Recommender Systems.” Directly relevant to selecting useful questions or attributes for preference elicitation.

He et al. (2024), “TUT4CRS: Time-aware User-preference Tracking for Conversational Recommendation System.” Relevant to tracking changing user preferences across conversation turns.

Ziegfeld et al. (2025), “The effect of preference elicitation methods on the user experience in conversational recommender systems.” Useful for understanding preference elicitation and interaction efficiency.

Kook et al. (2025), “Empowering Retrieval-based Conversational Recommendation with Contrasting User Preferences.” Useful for preference representation and recent implementation practice.

2025, “Stop Playing the Guessing Game! Evaluating Conversational Recommender Systems via Target-free User Simulation.” Relevant to evaluating preference elicitation and conversational recommendation without relying only on a known target item.

These papers form the initial reading set. Students should extend the literature review with relevant 2024–2026 work and papers cited by these studies.

3. Research Gap

Recent CRS research has studied dynamic preference tracking, preference elicitation, contrasting preferences, and strategies for selecting useful questions. Therefore, the project should not claim that these individual problems are unexplored.

The proposed study focuses on the interaction between these components: using the current conversational context and the user's available historical preference information to maintain a current preference state, estimate whether the system has enough information, and decide whether to elicit another preference or proceed to recommendation.

Research question: Can a context-aware current preference state, informed by historical user preferences and updated through preference elicitation, improve the selection of the next conversational action and recommendation quality?

The exact research gap and contribution will be finalized after the initial literature review and baseline comparison.

4. Proposed Methodology

The project will begin with an existing CRS baseline. The team will study how conversation history, user history, and preferences are represented in the selected implementation and then introduce a focused preference-state and adaptive-action component.

4.1 Tentative Architecture

The architecture is intentionally tentative. The exact representation, preference-elicitation method, action-selection strategy, and preference-fusion method will be finalized after the literature review and baseline reproduction.

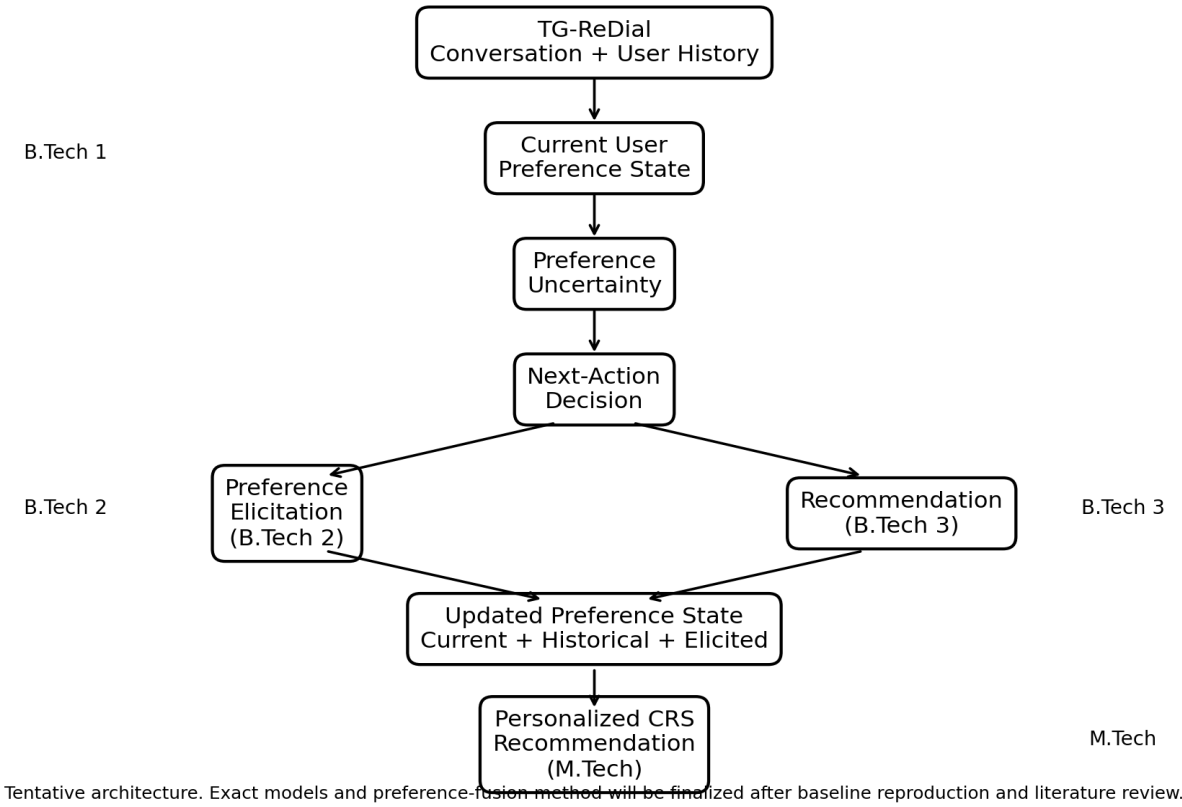


Figure 1. Tentative architecture and student-wise contribution.

Conversation Input. TG-ReDial multi-turn conversation together with available user profile and historical interaction information.

Current User Preference State. Represent the user's current interests, dislikes, constraints, and uncertainty using the conversation and available user history.

Preference Uncertainty. Estimate whether the current preference state contains enough information for the next action.

Next-Action Decision. Determine whether the system should request additional preference information or proceed toward recommendation.

Preference Elicitation. Select an informative attribute or question that can reduce uncertainty about the user's preferences.

Updated Preference State. Combine newly obtained information with the existing current and historical preference information.

Personalized CRS Recommendation. The M.Tech student investigates how current, historical, and elicited preferences can be combined before passing the resulting representation to an existing CRS recommendation mechanism.

5. Dataset and Existing Implementations

Primary dataset – TG-ReDial. TG-ReDial is preferred for Batch 2 because it contains multi-turn conversations together with user profiles and historical watching/interactions, making it better aligned with personalized preference research than using ReDial alone.

Secondary dataset – ReDial. ReDial can be used for additional validation if the main pipeline is stable by late October. It should not be introduced at the beginning of the project.

TG-ReDial. Dataset and code repository for the primary experiment.
https://github.com/RUCAIBox/TG-ReDial

CRSLab. Primary toolkit for establishing an existing CRS baseline.
https://github.com/RUCAIBox/CRSLab

UniCRS. Reference implementation for knowledge-enhanced conversational recommendation.
https://github.com/RUCAIBox/UniCRS

UserSimCRS. Optional resource for simulation-based CRS evaluation; not a mandatory dependency.
https://github.com/iai-group/UserSimCRS

CORAL. Recent preference-aware implementation useful for preference representation.
https://github.com/kookeej/CORAL

The preferred sequence is: reproduce a baseline; understand its data flow and evaluation; implement each research component separately; integrate the components; and then conduct controlled comparisons. No new conversational dataset should be created within this three-month project.

6. Division of the Task

B.Tech Student 1 – Current User Preference State

Study how the current conversation and available user history can be converted into a useful current preference state. Focus on interests, dislikes, constraints, and changes in preference. Establish the baseline data flow and prepare the preference-state representation.

B.Tech Student 2 – Preference Elicitation

Study what additional information should be requested when the current preference state is incomplete. Start with a simple information-gain, uncertainty-reduction, or candidate-space-reduction strategy before considering advanced approaches.

B.Tech Student 3 – Next-Action Decision

Study when the system should ask another question and when it should proceed toward recommendation. Start with a simple rule or threshold-based decision and evaluate its effect before attempting a learned policy.

M.Tech Student – Personalized Preference Fusion and CRS Integration

Own the central research methodology that connects the three B.Tech contributions. Investigate how historical user preferences, current conversational preferences, and newly elicited preferences can be combined into a personalized preference representation and used by the existing CRS recommendation mechanism. The M.Tech student will also lead the baseline/proposed comparison, ablation studies, experimental protocol, result analysis, and integration required for the journal paper.

7. Work Integration

The three B.Tech students work in parallel but contribute to one common pipeline. Student 1 prepares the current preference state, Student 2 studies how additional preference information can be elicited, and Student 3 studies the next-action decision. The M.Tech student connects these components through the personalized preference-fusion methodology and the existing CRS recommendation pipeline.

The M.Tech student's role is not limited to software integration. The M.Tech work constitutes the central methodological contribution of the journal paper: comparing current-only, historical-only, current-plus-historical, and full

current-plus-historical-plus-elicited preference representations.

Difficult examples and failure cases should be maintained in a shared experiment log so that the three B.Tech components remain consistent.

8. Experimental Design and Evaluation

The baseline will first be reproduced on TG-ReDial. The proposed approach will then introduce preference-state modeling, adaptive elicitation, and personalized preference fusion while keeping the remaining experimental setting as consistent as possible.

Baseline: Conversation + available user history → Existing CRS → Recommendation

Proposed: Conversation + user history → Current Preference State → Preference Uncertainty → Ask / Recommend → Updated Preference State → Personalized Preference Fusion → Existing CRS → Recommendation

Ablation studies should include, where feasible: current conversation only; historical preference only; current + historical preference fusion; and the full approach with elicited preference information. This provides a direct test of the M.Tech student's central contribution.

Evaluation Component	Metrics / Analysis
Recommendation quality	Recall@10, Recall@20, MRR, NDCG, or metrics supported by the selected baseline.
Preference elicitation	Information gain, uncertainty reduction, candidate-space reduction, or a suitable selection metric.
Interaction efficiency	Number of questions/turns required to reach a recommendation of a given quality.
Preference-state quality	Accuracy/F1 where reliable reference labels are available; otherwise downstream recommendation improvement and error analysis.
Error analysis	Unnecessary questions, missed preference changes, incorrect preference updates, and recommendation errors.

9. Tentative Timeline

Period	B.Tech 1	B.Tech 2	B.Tech 3	M.Tech
Week 1–2	Study TG-ReDial, CRS basics, user history and preference representation.	Study preference elicitation and identify candidate attributes/questions.	Study CRS recommendation and next-action decisions.	Study TG-ReDial, baseline CRS, personalization literature and define the central experimental framework.
Week 3–4	Run baseline and prepare preference-state data flow.	Prepare elicitation examples and a simple selection baseline.	Run recommendation baseline and define action-decision evaluation.	Reproduce baseline, establish evaluation pipeline, and define preference-fusion experiments.
Week 5–6	Develop current preference-state component.	Develop information-gain/uncertainty-reduction elicitation component.	Develop simple next-action decision component.	Develop initial current + historical preference-fusion method and integration plan.
Week 7–8	Refine preference state and analyze errors.	Evaluate elicitation quality and interaction efficiency.	Evaluate action decision and recommendation impact.	Run fusion experiments and compare current-only vs historical-aware representations.
Week 9–10	Support common pipeline integration.	Integrate elicitation.	Integrate next-action decision.	Complete personalized fusion, integration, ablations, and main experiments.
Week 11	Prepare preference-state analysis.	Prepare elicitation and interaction results.	Prepare action/recommendation analysis.	Lead final experiments, error analysis, tables, figures, and statistical comparisons.

Period	B.Tech 1	B.Tech 2	B.Tech 3	M.Tech
Week 12	Manuscript support and final checks.	Manuscript support and final checks.	Manuscript support and final checks.	Lead manuscript preparation and final submission package.

Milestones. A working TG-ReDial baseline by the end of September; individual components working by mid-October; preliminary integration and preference-fusion comparison by the end of October; final experiments, ablation studies, error analysis, and manuscript preparation during November.

10. Expected Outcome

The team is expected to develop and experimentally evaluate a personalized conversational recommendation approach that combines historical user preferences, current conversational preferences, and newly elicited preferences. The study should determine whether this personalized preference fusion improves recommendation quality and/or reduces the amount of interaction needed to reach a suitable recommendation.

The primary output is a research manuscript supported by reproducible experiments. A web application or user-facing software product is not part of the project requirement.

11. References and Initial Resources

Li, R., et al. (2018). Towards Deep Conversational Recommendations. NeurIPS.
https://papers.neurips.cc/paper/8180-towards-deep-conversational-recommendations

Zhou, K., et al. (2021). CRSLab: An Open-Source Toolkit for Building Conversational Recommender System. ACL-IJCNLP.
https://aclanthology.org/2021.acl-demo.22/

Makarova, A., et al. (2024). Learning a Strategy for Preference Elicitation in Conversational Recommender Systems.
https://centaur.reading.ac.uk/116044/

He, D., et al. (2024). TUT4CRS: Time-aware User-preference Tracking for Conversational Recommendation System.
https://openreview.net/forum?id=M4XBOWqXnP

Ziegfeld, L., Di Scala, D., & Cremers, A. H. M. (2025). The effect of preference elicitation methods on the user experience in conversational recommender systems.
https://doi.org/10.1016/j.csl.2024.101696

Kook, H., Kim, J., Park, S., & Lee, J. (2025). Empowering Retrieval-based Conversational Recommendation with Contrasting User Preferences. NAACL-HLT 2025.
https://aclanthology.org/2025.naacl-long.392/

2025. Stop Playing the Guessing Game! Evaluating Conversational Recommender Systems via Target-free User Simulation.
https://aclanthology.org/2025.findings-emnlp.1067/

TG-ReDial project repository and resources.
https://github.com/RUCAIBox/TG-ReDial

Research-team note. The architecture and individual methods are intentionally tentative. Students should not assume that a particular neural model, reinforcement-learning method, language model, or preference-fusion formula is mandatory. The final choices should be justified from the literature and finalized after the baseline is reproduced.